

Challenge 2: Can We Trust the Spam Filter?

Answer key

Q1 - Problem and metric

Spam is the positive class. A false positive is a legitimate email incorrectly flagged as spam, so an important message could be hidden. A false negative is a spam email incorrectly allowed through. Precision should guide the decision because it measures how trustworthy a spam flag is: among emails flagged as spam, how many are actually spam. Accuracy alone is insufficient because it treats false positives and false negatives equally and does not represent the stated cost of blocking legitimate mail. The mean cross-validation recall requirement of at least 0.80 prevents a trivial high-precision filter that flags very few messages and therefore misses too much spam.

Q2 - Gaussian Naive Bayes baseline

k	Mean precision	Precision SD	Mean recall	Recall SD
3	0.7113	0.0103	0.9576	0.0102
4	0.7117	0.0163	0.9531	0.0066
5	0.7127	0.0280	0.9553	0.0131

Actual / predicted	Legitimate (0)	Spam (1)
Legitimate (0)	1506	519
Spam (1)	60	1283

The five-fold matrix is an out-of-fold training matrix, not the final test matrix. Gaussian Naive Bayes catches most spam, but it also flags 519 legitimate messages as spam. That produces very high recall but much lower precision than desired for this scenario. The conclusion is stable across $k = 3, 4$, and 5 because the mean precision and recall values are very similar. This result should be interpreted as a limitation of this Gaussian Naive Bayes baseline on these numerical frequency features, not as evidence that Naive Bayes is generally unsuitable for spam filtering.

Q3 - Cross-validation, regularization, and overfitting

$C = 0.001$ is the most strongly regularized setting. The five-fold results are:

C	Training loss	CV validation loss	CV precision	CV recall
0.001	0.4086	0.4130	0.8985	0.8414
0.01	0.2751	0.2884	0.9183	0.8414
0.1	0.2209	0.2494	0.9180	0.8727
1	0.2041	0.2519	0.9127	0.8831
10	0.1983	0.2584	0.9160	0.8898
100	0.1965	0.2644	0.9114	0.8868
1000	0.1963	0.2703	0.9122	0.8891

At $C = 0.001$, both training and validation loss are high relative to the other settings, which is consistent with excessive regularization and underfitting. As C increases, training loss keeps decreasing. Validation loss improves at first, reaches its best five-fold value around $C = 0.1$, and then worsens as regularization becomes weaker. From $C = 0.1$ to $C = 1000$, the five-fold train-validation loss gap grows from about 0.029 to about 0.074, which is evidence of increasing overfitting. Precision is noisier and does not follow the loss curve perfectly.

The same broad pattern appears for $k = 3, 4$, and 5 , so the overfitting conclusion does not depend on one fold count. The exact preferred C is less stable, and fold variation is large enough that very small score differences should not be treated as decisive. The error bars represent fold standard deviation, not confidence intervals. Scaling must be fitted inside each fold so that the validation portion does not influence the mean and standard deviation used to transform the fitting data. Fitting the scaler before cross-validation would leak information from validation rows into training.

Q4 - Model selection before testing

Using the predeclared five-fold rule, choose $C = 0.01$ because it has the highest mean CV precision among settings with mean CV recall ≥ 0.80 . Its five-fold mean precision is 0.9183 and mean recall is 0.8414. $C = 0.1$ is almost tied, with precision 0.9180 and recall 0.8727. Both $k = 3$ and $k = 4$ instead favor $C = 0.1$ on precision, so the exact optimum is sensitive to the fold count and should not be described as decisive.

For the same five-fold comparison, Gaussian Naive Bayes has precision 0.7127 and recall 0.9553. The logistic model therefore sacrifices some spam detection in exchange for far more trustworthy spam flags. That tradeoff matches the stated problem, where incorrectly hiding legitimate email is the larger concern.

Q5 - Final unseen test evaluation

Actual / predicted	Legitimate (0)	Spam (1)
Legitimate (0)	485	21
Spam (1)	41	295

Accuracy	Precision	Recall	FP	FN
0.9264	0.9335	0.8780	21	41

There are 21 false positives, meaning 21 legitimate emails were incorrectly flagged as spam, and 41 false negatives, meaning 41 spam emails were allowed through. The test precision of 0.9335 and recall of 0.8780 are reasonably consistent with the five-fold CV estimates of 0.9183 precision and 0.8414 recall, allowing for sampling variation and the fact that the final model is refit on all training rows. The unseen test result therefore supports the recommendation. C, k, features, or thresholds must not be changed after viewing this result because doing so would use the test set for model selection and destroy its role as an independent final evaluation. The test score is an estimate, not a deployment guarantee.

Q6 - Reflection and limits

A useful Thinking Partner contribution was clarifying why StandardScaler belongs inside the model pipeline during cross-validation. The suggestion was checked against the experiment structure by verifying that each fold fits a fresh pipeline and therefore learns scaling parameters only from that fold's fitting rows before transforming the held-out rows. One limitation is that Spambase is a historical, corpus-specific email dataset and may not represent modern email traffic or current spam strategies. A useful next investigation would be repeated cross-validation on the training data, a validation-based threshold study, or evaluation on a contemporary dataset.